



Rudder Pedals

Electronics, Firmware & Configuration Guide

Applies to	Details
Device	Rudder Pedals with toe brakes
Firmware	HOTAS Rudder Pedals Universal Firmware (04_Rudder_Pedals.ino)
Configuration app	Tactical MFD Control Hub (Windows application, .exe)
Firmware updates	https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main
Companion document	Rudder Pedals – Assembly Instructions (axes layout, BOM, schematic, mechanical build)

Use this guide after the mechanical assembly is finished. It covers the sensor wiring, how to flash the firmware and how to configure the pedals with the Tactical MFD Control Hub app (calibration, axis inversion and filters).

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1.- How the system works

The pedals are the simplest device in the line: three SS49E hall sensors wired straight to three analog pins of an Arduino Pro Micro. There is no I2C bus and there are no buttons. The pedals show up in games as a controller with three axes.

Axis	Control	Sensor location	Arduino pin
Rudder	Pedals, left / right yaw	Main sensor mount, in the base	A0
Rx	Left toe brake	Left pedal arm	A1
Ry	Right toe brake	Right pedal arm	A2

1.1.- Where your settings are stored

Calibration, axis inversion, filters and the jitter threshold are stored in the Pro Micro's internal EEPROM. They survive power cycles and firmware updates.

2.- Wiring

Sensor pin	Connects to
OUT – rudder sensor	Pro Micro A0
OUT – left toe brake sensor	Pro Micro A1
OUT – right toe brake sensor	Pro Micro A2
VCC of all three sensors	Pro Micro VCC (5 V)
GND of all three sensors	Pro Micro GND

SS49E pinout

Looking at the sensor's printed face with the legs pointing down, the pins are, from left to right: **VCC, GND, OUT**. Swapping VCC and GND can destroy the sensor, so check twice before powering up.

2.1.- Quick sensor check

After flashing the firmware (section 3), open the app's MON > INPUT page (section 4.6) and move each control: the matching axis must move smoothly from one end to the other. An axis that doesn't move, or jumps to one end, points to a wiring problem or a magnet that is too far from its sensor.

3.- Firmware installation and updates

3.1.- Required libraries

Header	Library	How to install
Joystick.h	ArduinoJoystickLibrary (Matthew Heironimus)	Download the ZIP from GitHub → Sketch → Include Library → Add .ZIP Library
EEPROM.h	Built in	—

3.2.- Custom USB name and VID/PID (do this before the first upload)

By default the pedals show up in Windows and in games as "Arduino Leonardo". Giving them their own name and USB IDs makes them easy to recognize, and keeps games from mixing up their bindings with other Arduino-based devices. Set them **before** uploading, so you only flash once.

1. Find the AVR core folder that contains `boards.txt`, usually `%LOCALAPPDATA%\Arduino15\packages\arduino\hardware\avr\1.x.x\` (the exact path may vary).
2. In that same folder, create a text file named `boards.local.txt` with the lines below. Values in this file override `boards.txt` without editing it.
3. Restart the Arduino IDE.

```
leonardo.build.vid=0x2342
leonardo.build.pid=0x803B
leonardo.build.usb_product="Rudder Pedals"
```

If you also build other Noisy Boi devices

`boards.local.txt` applies to every Leonardo / Pro Micro you upload from this PC. Each device needs its own PID (for example, 0x8038 for the joystick, 0x8039 for the throttle, 0x803A for the collective and 0x803B for the pedals), so edit the file before flashing a different device.

3.3.- Upload the firmware

1. Open `04_Rudder_Pedals.ino`. Always use the latest version from the FIRMWARE folder of the official repository (section 3.5).
2. Select **Tools** → **Board** → **Arduino Leonardo** and the pedals' port.
3. Click **Upload**.
4. If the upload fails because the port disappears, short **RST** to **GND** on the Pro Micro and click Upload right away (some clones need two quick taps).

After the upload, the pedals show up in Windows as a game controller with three axes **and** as a serial (COM) port. The app uses that port. You no longer need to edit the code with calibration values, or remove the serial code after calibrating.

3.4.- Factory defaults

Setting	Default
Inverted axes	Rudder, Rx and Ry (all three)
Alpha 1 (rudder)	0.30
Alpha 2 (toe brakes)	0.20
Jitter threshold	15
Calibration	Generic values; calibrate before flying (section 4.5)

3.5.- Firmware updates

Firmware updates, fixes and new features are published in the official GitHub repository:

<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main>

1. Open the repository and download the latest pedals firmware from the FIRMWARE folder (**Code** → **Download ZIP**, or open the file and use **Download raw file**).
2. Check the repository notes for changes, new required libraries or new steps before updating.
3. Close the Tactical MFD Control Hub (it uses the same COM port), then upload the new `04_Rudder_Pedals.ino` exactly as in section 3.3. Check that `boards.local.txt` still has the pedals' PID.
4. Reconnect with the app, press **STATUS** on the LINK page to see the installed version in the log (line Firmware: ...), and check CTRL > INV / FILT and MON > INPUT.

Your settings are kept

Calibration and settings live in EEPROM, so a normal update keeps them. Recalibrate only if the update notes ask for it, or if the axes behave differently afterwards.

Watch the repository

On GitHub, click **Watch** → **Custom** → **Releases** to get an email when a new version is published.

4.- Tactical MFD Control Hub (configuration app)

The Tactical MFD Control Hub is a Windows desktop app styled as a fighter-jet multifunction display (MFD). With the pedals you can use it to:

- calibrate the rudder and both toe brakes,
- invert axes and tune the filters,
- check the three axes live.

All settings are stored on the pedals themselves. **You do not need the app running to fly.**

4.1.- Installation and launch

The app is a single Windows program (Windows 10 or 11, 64-bit). It needs no installer and no additional software.

1. Download Tactical MFD Control Hub.exe from the **Releases** page of the official repository (<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/releases>) and save it in any folder, for example Documents\HOTAS. You can create a desktop shortcut to it.
2. Double-click it to start. The first start can take a few seconds while Windows unpacks the program.
3. If Windows shows "**Windows protected your PC**", click **More info** → **Run anyway**. The app is not signed with a paid code-signing certificate, so Windows SmartScreen warns about any new program. This happens only the first time.
4. If your antivirus quarantines the file, restore it and add an exception for it. Small independent .exe tools are sometimes flagged as a false positive.

Before connecting

Only one program can use the COM port at a time. **Close the Arduino IDE Serial Monitor** (and any other serial tool) before connecting from the app.

4.2.- Navigating the MFD

Like a real MFD, the screen is surrounded by Option Select Buttons (OSBs). The label next to each OSB shows what it does on the current page. You can click the OSBs, or click directly on the rows, toggles and sliders shown on the screen.

OSB group	Function
Bottom row	Page selection: LINK , CTRL , MON . The active page is boxed.
Left column	Sub-pages (or the port list on LINK).
Right column	Page actions. On CTRL: UP / DN move the cursor, - / + change a slider, SEL toggles or runs the selected row.

The **LINK** flag in the header strip lights up while the app is connected. The GUN, MSL, TD and HFB flags belong to the joystick's haptic features and stay off with the pedals.

4.3.- LINK page – connecting to the pedals



LINK page: port list, connection state and serial log

1. Plug in the pedals and press **SCAN** (bottom-left OSB) to refresh the port list.
2. Select the pedals' COM port with the left OSBs (or click it in the list).
3. Press **CONN**. The app requests the device status.
4. When the pedals answer, **DEVICE** shows **RUDDER PEDALS** and the app opens the CTRL page automatically.
 - **STATUS** asks the pedals for their current settings again.
 - **DISC** closes the connection.
 - Not sure which port it is? Note the list, unplug the pedals, press SCAN, and see which port disappeared.

4.4.- CTRL page – pedal settings

Choose a sub-page with the left OSBs: **INV**, **FILT**, **CAL**. Every change is sent to the pedals and saved immediately; the app then re-reads the status to confirm. The pedals have no special modes (MODE shows "NO SPECIAL MODES FOR THIS DEVICE") and no haptics (HFB is for the joystick only).

INV – Axis inversion



CTRL > INV (factory default: all three inverted)

Toggle **INVERT RUDDER**, **INVERT RX** (left toe brake) or **INVERT RY** (right toe brake) until each control moves the right way in-game. All three are inverted by default, which matches the standard build; change them only if an axis moves backwards.

FILT – Filters



CTRL > FILT

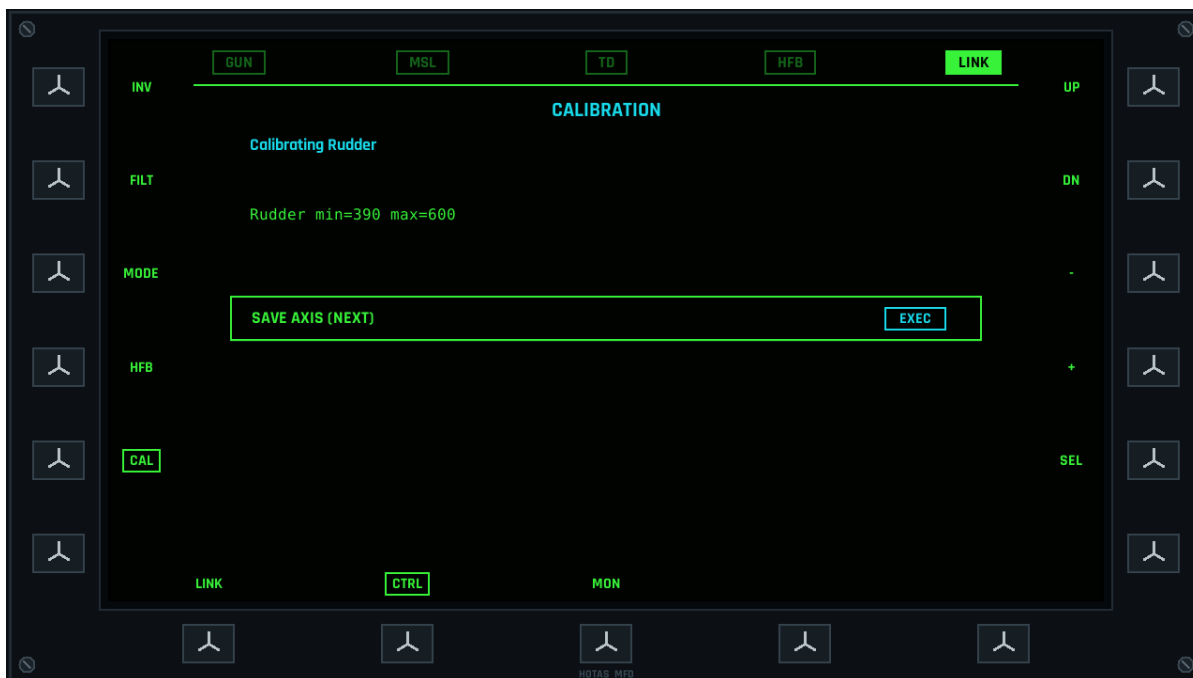
Setting	Range	Default	Effect
ALPHA 1 (RUDDER)	0.01 – 1.00	0.30	Rudder filter. Lower = smoother but slower to respond. Higher = more direct, more noise.
ALPHA 2 (TOE BRAKES)	0.01 – 1.00	0.20	Same, for both toe brakes.
JITTER THRESHOLD	0 – 500	15	Minimum change (output range ± 16384) before an axis value updates. Raise it if an axis flickers at rest.

Fast movements are not slowed down: when a pedal moves quickly the filter adapts and follows it almost instantly. Drag a slider and release it, or use **- / +**; the value is sent when you release.

4.5.- CAL – Calibration

Calibration teaches the firmware the real minimum and maximum of each sensor, so every axis uses its full range. Calibrate once after the build, and again if you move a sensor, a magnet or change the springs.

1. Connect on the LINK page, then go to **CTRL > CAL**.
2. Select **START CALIBRATION WIZARD**. The pedals start with the rudder right away.
3. **Rudder**: push the left pedal fully forward, then the right pedal fully forward, several times, until the **min / max** values stop changing. Select **SAVE AXIS (NEXT)**.
4. **Rx – left toe brake**: press it fully and release it, several times. Select **SAVE AXIS (NEXT)**.
5. **Ry – right toe brake**: same as the left one. Select **SAVE AXIS (NEXT)**.
6. The screen shows **CALIBRATION COMPLETE!** and the values are stored.



Calibrating the rudder: live min/max values

Important

The app refuses to save a range that is too small (less than 20 counts): move the control through its full travel and try again.

The rudder's center is the midpoint between the two calibrated ends. If it doesn't read close to zero with your feet off the pedals, the sensor or magnet is off-center: adjust the main sensor mount, or add a small deadzone in the game.

While the wizard runs, the pedals stop sending inputs to the PC. Finish it before flying. If the app is closed in the middle of a calibration, the pedals keep waiting. Reopen the app and reconnect (it picks up the wizard where it was), or unplug and replug the USB cable to abort.

4.6.- MON page – monitoring

INPUT – Pedal inputs



MON > INPUT: the three axes

The three axes are shown live. Use **JOY-** / **JOY+** to choose the pedals and **SCAN** to refresh the list. Move each control on its own to see which bar belongs to it.

STAT – Status summary



MON > STAT

A one-page summary of the connection and the device settings. Handy when you ask for support: send a screenshot of this page.

5.- Troubleshooting

Symptom	What to check
App: "Failed to connect"	Another program has the port open (Arduino Serial Monitor, another app instance). Close it and press CONN again.
Connected, but DEVICE stays "----"	Press STATUS. If nothing appears in the log, re-flash the firmware and check that you selected the pedals' COM port.
An axis doesn't move	Check that sensor's OUT wire (A0, A1 or A2) and its VCC/GND, and the distance to its magnet.
Two axes are swapped	Compare the wiring with the table in section 2.
Rudder isn't centered at rest	The sensor or magnet is off-center: adjust the main sensor mount and recalibrate.
A toe brake feels coarse or steppy	Its magnet barely moves relative to the sensor. Check the 5×2 mm magnets and the stopper, then recalibrate using the full travel.
An axis jitters at rest	Raise the JITTER THRESHOLD, or lower the corresponding ALPHA.
Axis moves the wrong way	CTRL > INV.
Inputs freeze in-game	A calibration is still running on the pedals: finish it, or replug the USB cable.
Games confuse the pedals with another device	Give them their own PID (section 3.2) and rebind them in the game.

Appendix A.- Serial command reference

Everything the app does can also be done from any serial terminal (for example, the Arduino Serial Monitor at **115200** baud with **Newline** line ending). Commands are not case-sensitive.

Command	Action
STATUS	Print the current settings
CAL	Start the calibration: rudder, left toe brake, right toe brake. For each axis, move it end to end and press Enter (empty line) to save.

Command	Action
INV_RUDDER, INV_RX, INV_RY	Toggle the inversion of that axis
FIL1 <0.01–1.00>	Set Alpha 1 (rudder filter)
FIL2 <0.01–1.00>	Set Alpha 2 (toe brakes filter)
JITTER <0–500>	Set the jitter threshold

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ALL DONE – HAPPY FLYING!